TIRTHA PATIL

B.Tech. CSE (Artificial Intelligence & Machine Learning) | AI/ML • Computer Vision • XAI

Email: patiltirtha8811@gmail.com | Phone: +91 8446125185 | Personal Portfolio | GitHub | LinkedIn

EDUCATION

B.Tech. Computer Science & Engineering (Artificial Intelligence & Machine Learning)
Vishwakarma Institute of Technology, Pune | Expected Graduation: 30 June 2028 | CGPA: 8.22/10

- Coursework: Machine Learning, Artificial Intelligence, Data Structures, DBMS, Operating Systems, Probability & Statistics, Linear Algebra

TECHNICAL SKILLS

- **Programming:** Python, C++, HTML/CSS, JavaScript
- **Machine Learning:** Scikit-learn, CNN, Transfer Learning, Random Forest, SVM, SHAP, Grad-CAM
- **Libraries:** NumPy, Pandas, Matplotlib, Seaborn, OpenCV
- **Tools & Frameworks:** FastAPI, Git/GitHub, Jupyter Notebook, VS Code

EXPERIENCE

Infosys Springboard Virtual Internship 6.0 | Python Project Intern (certified)

- Collaborated in a six-member team to develop Chatterbox, a real-time chat application.
- The team was selected to present the project to Infosys professionals.

PROJECTS

Retina Disease Detection using CNN & Grad-CAM (Jun 2025 – Dec 2025) GitHub

- Developed a 3-class retinal disease classification model using MobileNetV2, TensorFlow, and Keras on APTOS 2019 and G1020 datasets.
- Implemented Grad-CAM heatmaps and SHAP-based explainability to interpret model predictions for Diabetic Retinopathy, Glaucoma, and Normal retinal images.

Lifestyle-Based Health Risk Prediction (Jan 2026 – Jun 2026) GitHub

- Built and compared multiple ML classifiers, including Gradient Boosting, XGBoost, CatBoost, and Random Forest, achieving 75.18% accuracy with Gradient Boosting.
- Applied SHAP for feature-level interpretability and used Pandas, NumPy, and Scikit-learn for data preprocessing and model evaluation.

Chatterbox: Real-Time Chat Application (Internship Project) Live Demo | GitHub

- Developed a real-time messaging application using Python, FastAPI, Uvicorn, WebSockets, and Socket.IO for bidirectional communication.
- Implemented an asynchronous backend architecture to support live message exchange and event-driven communication.

Space Rover Exploration Agent (Jan 2026 – Jun 2026) GitHub

- Developed an intelligent rover navigation agent in Python implementing Breadth-First Search (BFS), Depth-First Search (DFS), and A* to navigate terrain and find optimal paths.

Medical Data Visualization (Personal) GitHub

- Performed exploratory data analysis on the Pima Indians Diabetes Database using Python, Pandas, NumPy, Matplotlib, and Seaborn.
- Analyzed feature distributions and correlation matrices, identifying glucose as a prominent feature associated with diabetes outcomes.

Diabetes Outcome Prediction using Machine Learning (Personal) GitHub

- Developed a supervised ML pipeline using the Pima Indians Diabetes Database to predict diabetes outcomes from clinical health indicators.
- Applied data preprocessing, feature analysis, and model evaluation using Python and Scikit-learn.

OS Page Replacement Simulator (Jan 2026 – Jun 2026) GitHub

- Implemented FIFO, LRU, and Optimal Page Replacement algorithms to simulate memory management and page fault handling in Python/C.
- Compared page fault counts across different reference strings and frame capacities to evaluate algorithm efficiency and memory utilization.

Personal Portfolio Website (Personal) Live Demo | GitHub

- Designed and developed a responsive personal portfolio website to showcase AI/ML projects, technical skills, research interests, and professional achievements.

Books Shopping Website (Personal) GitHub

- Developed a web-based bookstore that allows users to browse books, explore categories, view book details, and shop through an intuitive interface.
- Built using HTML, CSS, and JavaScript, with a focus on responsive design and user-friendly navigation.

AeroSync Flight Attendance Booking and Management System (Jun 2025 – Dec 2025) GitHub

- Developed a flight management system using C supporting flight scheduling, passenger booking, attendance tracking, seat allocation, and file-based data persistence.
- Built a web dashboard using HTML, CSS, and JavaScript with role-based access, fare calculation, and flight occupancy analytics.

CERTIFICATIONS

Programs & Simulations

McKinsey.org Forward Program Dec 2025

- Completed practical training in structured problem-solving, communication, adaptability, and digital skills.

Deloitte Cyber Job Simulation Jul 2025

- Completed practical cybersecurity tasks simulating real-world professional scenarios through Forage.

1 Million Prompters Initiative

- Earned a certificate focused on prompt engineering and effective use of AI systems.

Courses & Specializations

Deep Learning Specialization | DeepLearning.AI, Coursera Oct 2026

- Completed five courses covering neural networks, CNNs, RNNs, LSTMs, Transformers, and deep learning model optimization using Python and TensorFlow.

Financial Technology (Fintech) Innovations | University of Michigan, Coursera Oct 2026

- Completed a four-course specialization in payment technologies, blockchain, cryptocurrency, crowdfunding, and AI-driven investment.

Infosys Springboard

Certifications in:

- Python Foundations
- Programming Fundamentals Using Python (Parts 1 & 2)
- Object-Oriented Programming Using Python
- Basics of Python
- Database Management Systems (Parts 1 & 2)
- Introduction to NoSQL Databases
- Software Engineering and Agile Software Development
- Agile Scrum in Practice

EXTRACURRICULAR

French Language:

- DELF A1 and A2 certified
- Currently pursuing B1

Campus Involvement:

- Video Editing Coordinator at Abhivriddhi (Student Development Committee)
- Organized a Resin Art Workshop with Student Council at VIT, Pune